



9645 - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Tommaso L. Treu (PI)	University of California - Los Angeles
Dr. Guido Roberts-Borsani (CoI) (ESA Member) (CoPI) (Contact)	University College London
Dr. Adriano Fontana (CoI) (ESA Member) (CoPI) (Contact)	INAF - Osservatorio Astronomico di Roma
Prof. Richard S. Ellis (CoI) (ESA Member)	University College London
Dr. Eros Vanzella (CoI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Dr. Takahiro Morishita (CoI)	Astronomical Institute, Tohoku University
Dr. Laura Pentericci (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Pietro Bergamini (CoI) (ESA Member)	Universita di Milano
Antonello Calabro' (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Marco Castellano (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Prof. Claudio Grillo (CoI) (ESA Member)	Universita di Milano
Prof. Tucker Jones (CoI)	University of California - Davis
Prof. Matthew A. Malkan (CoI)	University of California - Los Angeles
Dr. Piero Rosati (CoI) (ESA Member)	Universita degli studi di Ferrara
Dr. Elena Gallo (CoI)	University of Michigan
Dr. Themiya Nanayakkara (CoI)	Swinburne University of Technology
Dr. Massimo Stiavelli (CoI)	Space Telescope Science Institute
Prof. Marusa Bradac (CoI)	University of Ljubljana, Dept. of Physics
Dr. Alan Dressler (CoI)	Carnegie Institution of Washington
Prof. Karl Glazebrook (CoI)	Swinburne University of Technology
Dr. Paola Santini (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma

<i>Name</i>	<i>Institution</i>
Prof. Michele Trenti (CoI)	University of Melbourne
Mr. Evan Haze Nunez (CoI)	California Institute of Technology
Dra. Sofia Rojas (CoI)	University of California - Los Angeles
Dr. Massimo Pascale (CoI)	University of California - Los Angeles
Dr. Davide Bevacqua (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Giovanni Gandolfi (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Borja Perez Diaz (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Mr. Raunaq Singh Rai (CoI) (ESA Member)	University College London

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSpec_Faint_Deep				
	1	Grating_Faint_DiagW_v7_AllSpec_Cy5	NIRSpec MultiObject Spectroscopy	(1) msa_targets_DiagWeight

ABSTRACT

What were the physical processes that governed star formation and the build up of metal and dust in the early universe? Did supermassive black holes provide a significant contribution to cosmic reionization? These major questions are emerging after the JWST discovery of surprisingly bright galaxies and AGNs in the reionization era. However, these hyperluminous galaxies are likely to be special (the "tip of the iceberg"), e.g. caught right after a burst or powered by accretion onto supermassive black holes. Comprehensive answers necessitate detailed studies of the dominant population of sub-luminous ($<L^*$) galaxies. This is only now possible by combining the unique power of JWST with strong gravitational lensing magnification.

We propose to carry out the first measurements of the ionizing radiation field, star formation history, dust content, and detailed chemical enrichment for the sub- L^* ($-18.5 < \text{MUV} < -14$, i.e. $0.002\text{--}0.1 L^*$) population at $z > 6$. This goal will be achieved via a $R \sim 1000$, 1-to-5 micron ultradeep spectroscopic survey of >120 faint sources at $z > 6$, magnified by the best-characterized lensing cluster, Abell 2744. These spectra will detect UV and optical emission line diagnostics that trace C,N,O abundance, escape fraction, and AGN activity.

Raw data will be public immediately. Science-grade data and catalogs will be delivered 6 months after observation, enabling a wide range of scientific investigations. Between the primary and ancillary science, this program will have a broad impact in astronomy, with general implications for galaxy formation, feedback, reionization, star formation, chemical enrichment, and black hole seeds and accretion.

OBSERVING DESCRIPTION

We will obtain ultradeep NIRSPEC spectroscopy in three setups: G140M-F100LP, G235MF170LP, and G395M-F290LP. This is achieved adopting a NRSIRS2 readout pattern and 3-shutter dither pattern. with a single highly optimized MSA design we will target $>120\ z>6$ galaxies with full exposure time. Additional $\sim 40\text{-}50$ galaxies will be included in the MSA configuration with partial shutters exposure.

Proposal 9645 - Targets - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	msa_targets_DiagWeight	RA: 00 14 19.5028 (3.5812617d) Dec: -30 23 29.91 (-30.39164d) Equinox: J2000		
	Comments: Description=[]				

Proposal 9645 - Observation 1 - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

Observation	Proposal 9645, Observation 1: Grating_Faint_DiagW_v7_AllSpec_Cy5 Fri Mar 13 17:02:20 GMT 2026 Diagnostic Status: Warning Observing Template: NIRSpec MultiObject Spectroscopy
-------------	---

Proposal 9645 - Observation 1 - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

[illegible]

Proposal 9645 - Observation 1 - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	msa_targets_DiagWeight	RA: 00 14 19.5028 (3.5812617d) Dec: -30 23 29.91 (-30.39164d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	4		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	5		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	6		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	7		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold		
	MSATA	false	No	MSA Center	Primary (79 sources)	Secondary (156 sources)	jwst-nirspec-mr		1.5		
Reference Stars											

Proposal 9645 - Observation 1 - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	2	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	3	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	4	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	5	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	6	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	7	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	8	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	9	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	10	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	11	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	12	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	15405.868
	13	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
	14	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201

Proposal 9645 - Observation 1 - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
15	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
16	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
17	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
18	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
19	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
20	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
21	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
22	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
23	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	6	14968.201
24	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
25	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
26	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
27	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
28	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801

Proposal 9645 - Observation 1 - The Core of the Iceberg: the Physics of Sub-L* Galaxies in the First Billion Years

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
29	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
30	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
31	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
32	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
33	3 (G395M/F290LP)	c1	3 Shutter Slitlet	3.5892193333333 333 Degrees - 30.376968055555 55 Degrees	39.996001886040 83			3	9	13917.801
Special Requirements	Group Visits within 53.0 Days									
	Visits Same PA MSA Planned Aperture PA 40.0000 to 40.0000 Degrees (V3 261.4254303 to 261.4254303)									